

# Beatless: A Deep Learning Approach for Percussive Removal in Music

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**Abstract**—Melody, harmony and rhythm form the music audio's composition with rhythm being defined as the sequence of beats and other percussive elements. Despite being a fundamental component of all styles of music, the presence of a beat can adversely impact many functions, such as karaoke generation, remixing, melody separation and transcribing music. Current methods for separating the harmonic and percussive aspects of audio (e.g., harmonic-percussive source separation, NMF), are not ideal options for reducing beats. The performance, quality, and applicability of these methods, when applied to different musical styles, is often poor.

With the rise of deep learning, music source separation has experienced major progress. Models such as U-Net, Wave-U-Net, and Demucs have shown great ability to extract and remove audio sources with high quality. This paper reviews recent progress in deep learning applied to beat removal, mainly focusing on innovations in model architectures, lightweight real-time models and rhythm-aware designs. It also presents exploratory data analysis (EDA), which shows some key audio features such as zero-crossing rate, spectral centroid, and tempo that differentiate characteristics of beats compared to melodic elements. Also, the paper identifies some research gaps, including the lack of beat-annotated datasets, insufficient evaluation metrics specific to beat suppression and limited real-time implementations. The paper concludes with future research directions for more robust, advanced and user-friendly beat removal systems.

**Keywords** - Music Source Separation, Beat Removal, Deep Learning, Spectrogram, U-Net, Demucs, Music Processing

## I. INTRODUCTION

In general, beats are the primary rhythmic aspects of a piece of music that exist alongside harmonic and melodic aspects, with many times being a critical input into rhythm. However, they are not always beneficial for various uses, e.g., when generating karaoke, transcoding, and when needing to isolate the melodic element cleanly. Many of the older methods of removing beats from music were produced through manual filtering and/or the use of Harmonic-Percussive Source Separation (HPSS) [18], but both of these methods can create unwanted artifacts in audio files and subsequently degrade overall audio performance. With the rapid expansion of the field of Deep Learning, it is now possible to implement advanced data driven learning algorithms that analyze complex audio features in musical recordings. CNNs, RNNs, and hybrid models, such as Demucs [4] and Wave-U-Net [3], have demonstrated superior capabilities in performing source separation on music recordings. This paper gives a focused review of deep learning techniques applied to beat removal in music. The problem this paper addresses is the absence of beat-specific

deep learning solutions in the current world that suppress percussive components while preserving melodic clarity. It also explores the evolution of advanced model architectures, evaluates existing tools and datasets, and shows some key findings from exploratory data analysis (EDA). Also, it identifies current limitations in the field and proposes future research directions focused at developing robust, real-time, and genre-specific beat removal systems.

## II. LITERATURE SURVEY

### A. Rise of Deep Learning in Music Source Separation

The rise of deep learning has greatly enhanced music source separation, which allows models to easily learn the mapping between mixed audio inputs and their individual stems of music. The U-Net architecture [2], primarily developed for biomedical image segmentation, was adapted for music separation tasks using spectrograms. As shown in Fig.1, its encoder-decoder structure with skip connections enables efficient feature extraction and accurate separation of harmonic and percussive elements.

The Wave-U-Net model [3] extends this concept to the time domain by using raw waveforms directly, thereby improving temporal fidelity at the cost of higher computational load. Demucs [4] is a hybrid model which combines convolutional and bidirectional LSTM layers, processes raw audio to record both local spectral features and long-term temporal patterns. Together, these models represent a significant shift from manual signal processing to data-driven learning, achieving great results in music and beat separation.

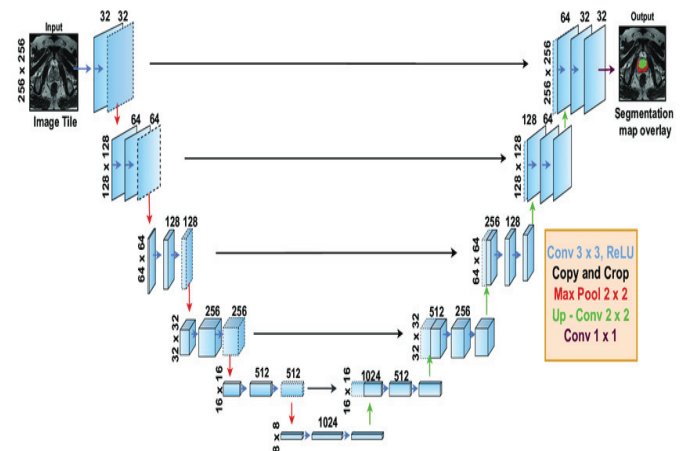


Fig. 1. Standard U-Net architecture for encoder-decoder based learning.

### B. Recent Model Innovations and Lightweight Architectures

Several recent papers propose some upgradations to improve separation quality, reduce model complexity and enable real-time model processing [5–10]. The Band-Split RoPE Transformer [5] uses a transformer architecture with rotary positional encoding in frequency bands, which results in state-of-the-art SDRs in drum separation tasks. Hybrid Y-Net [7] and DTTNet [6] use time–frequency cross-domain processing to enhance efficiency while retaining separation performance, making them usable for edge devices and mobile applications. Contrastive Latent Masking [9] applies contrastive learning to easily differentiate between source types, improving model generalization. Spectrogram-TasNet [10] is designed for real-time source separation by integrating TasNet’s time-domain processing technique with spectrogram-based learning.

### C. Beat and Rhythm-Aware Techniques

Early approaches mainly relied on signal processing methods. Vincent et al. [15] introduced objective performance metrics (SDR, SIR, SAR) that became the foundation for evaluating quality of separation. Fitzgerald et al. [18] demonstrated harmonic-percussive source separation (HPSS) using median filtering, a widely used baseline method. Rafii and Pardo [16] proposed the use of similarity matrices for voice separation, while Rafii et al. [17] later provided a comprehensive review of music source separation techniques. Bittner et al. [19] released MedleyDB, a multitrack dataset that enabled supervised training for separation of models in music.

The rise of deep learning transformed the field. Jansson et al. [2] adapted the U-Net architecture for singing voice separation using spectrograms, demonstrating the viability of encoder–decoder models. Stoller et al. [3] introduced Wave-U-Net, a fully convolutional model working directly on waveforms, improving time-domain fidelity. Défossez et al. [4] presented Demucs, combining CNNs and LSTMs to achieve state-of-the-art waveform-based separation. Luo and Mesgarani [8] proposed Conv-TasNet, a time-domain architecture initially designed for speech separation but later extended to music.

Recent works focused on lightweight and efficient architectures. Chen et al. [6] developed DTTNet, a dual-path UNet optimized for edge devices. Fernando et al. [7] introduced Hybrid Y-Net, combining spectrogram and waveform features for efficient singing voice separation. Kim et al. [9] applied contrastive learning with deep latent masking to improve model generalization. Schlüter [20] developed Open-Unmix, an open-source reference implementation for reproducible separation research.

Transformers have also been applied to this domain. Liu et al. [5] has used Band-Split RoPE Transformer, achieving state-of-the-art SDR for drum separation by modeling frequency bands with rotary positional encoding. Kong et al. [13] used transformer-based techniques for sound event detection, showing their promise for music related applications. Rafii et al. [1] introduced the MUSDB18 dataset, which has become the standard benchmark for music source separation. McFee et al. [14] developed Librosa, a widely used Python library for audio analysis which supports some feature extractions like spectrograms and tempo computation.

From this review, it is clearly evident that while deep learning models like U-Net [2], Wave-U-Net [3] and Demucs [4] show great performance in source separation, their application to beat-specific suppression remains limited. As represented in Table 1, the comparison shows the strengths and limitations of different deep learning architectures. This shows the need for beat-annotated datasets, specialized models, and evaluation metrics focused on rhythm and beat specific elements.

TABLE I. COMPARISON OF TECHNIQUES

| Model            | Input Type         | Strengths                             | Limitations                             |
|------------------|--------------------|---------------------------------------|-----------------------------------------|
| U-Net            | Spectrogram        | Simple and effective, well-studied    | Loses time-domain detail                |
| Wave-U-Net       | Waveform           | High-fidelity output                  | High compute, longer training           |
| Demucs           | Waveform           | Best-in-class quality                 | Requires significant data and training  |
| DTTNet/Y-Net     | Spectrogram + Wave | Efficient and lightweight             | May trade off quality on complex tracks |
| RoPE Transformer | Spectrogram        | Captures frequency band relationships | Complex and hard to train               |

### III. RESEARCH GAPS

- **Lack of Beat-Specific Model Training:** Current models are usually restricted to vocal or accompaniment separation and thus do not have a specific emphasis on the removal of only percussion. The absence of explicit training regarding beat-specific training limits models from achieving optimal and effective methods for the accurate removal of percussion from a musical piece.
- **Insufficient Evaluation Metrics:** SDR, SIR and SAR [15] are the most widely used evaluation metrics for offering an overall evaluation of source separation and as a result, these metrics offer inadequate information on the quality or effectiveness of a model's ability to remove percussion specifically.
- **Poor Genre Generalization:** Although many models have the ability to perform substantially well in terms of separating major genre tracks, but few are able to sufficiently separate percussion in more complex musical styles like electronic, hip-hop or jazz which shows major limitations of generalisation across genres.
- **Real-Time Application Gap:** While some models like Conv-TasNet [8] and Spectrogram -TasNet [10] are optimized for low-latency processing, very few are tuned for real-time beat suppression, which is crucial for live remixing, DJing, or performance settings.
- **User-Controlled Beat Removal is Unexplored:** Current models offer binary beat removal. There's limited research on models that allow adaptive beat suppression, e.g., partial beat attenuation based on user input or application context.

#### IV. PROPOSED METHODOLOGY

The proposed methodology for beat removal using deep learning is organized into five stages: data preprocessing, feature extraction, exploratory data analysis, model preparation and training, and evaluation with deployment considerations. Fig. 2 illustrates the five-stage workflow of the proposed methodology.

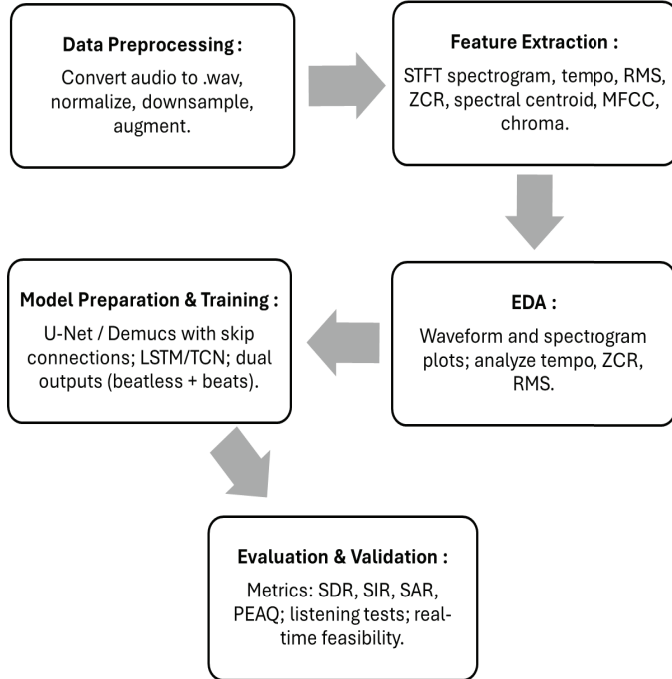


Fig. 2. Workflow Architecture of the proposed methodology

##### A. Data Preprocessing

The MUSDB18 dataset [1] is used as the main source of audio tracks in music source separation tasks. As all the files are originally in .stem.mp4 format, they are converted to .wav format for compatibility. Audio is normalized and down-sampled to ensure consistency across samples. To improve training efficiency, tracks are divided into short clips. Also, Data augmentation techniques such as pitch shifting, tempo scaling, and noise injection are used to increase model robustness and for better generalization across different music genres.

##### B. Feature Extraction

Time–frequency features are extracted using Short-Time Fourier Transform (STFT) to generate spectrograms. Also, Mel-frequency cepstral coefficients (MFCCs), chroma features, and onset strength envelopes are extracted to capture rhythmic and beat-related characteristics. Zero-Crossing Rate (ZCR), Root Mean Square (RMS) energy and spectral centroid are calculated to differentiate percussive sounds from harmonic components. All these features are very helpful for the model to learn both harmonic and rhythmic structures in the audio.

##### C. Exploratory Data Analysis (EDA)

Exploratory Data Analysis gives insight into how to characterize percussion and melody, and the visual representation of this information. The approach uses both waveform and spectrograms to view transients on waveforms, and to view vertical and horizontal groups of

sound on the spectrograms for the harmonic and percussive components. Using characteristic features like Tempo, Zero Crossing Rate (ZCR) and Root Mean Square (RMS), the differences between beats and non-beat sounds can be examined using the analysis of these differences.

##### D. Model Preparation and Training

The main part of methodology is built around a U-Net-based encoder–decoder architecture with skip connections. The encoder (contracting path) gets the hierarchical feature representations, while the decoder (expansive path) reconstructs the output signal. A bottleneck layer enhanced with LSTM/TCN units is included to record some long-range rhythmic dependencies. The model is trained on paired data. Which comprises of input mixture spectrogram and its target beatless spectrogram (mixture minus drums). Dual outputs are used: beatless audio as the main target and isolated beats as a secondary branch to improve learning of percussive sounds. Experiments also compare U-Net [2], Wave-U-Net [3], and Demucs [4] architectures.

##### E. Evaluation and Deployment

Evaluation is done using objective metrics like Signal-to-Distortion Ratio (SDR), Signal-to-Interference Ratio (SIR), and Signal-to-Artifact Ratio (SAR) [15]. Perceptual metrics such as PEAQ and Mean Opinion Score (MOS) from listening tests are used to check audio quality of the project. Real-time feasibility is explored by using lightweight models (DTTNet or quantized models). This system is designed to produce beat-suppressed audio suitable for karaoke and music analysis applications, with option of providing isolated beats for remixing or rhythm based tasks.

#### V. IMPLEMENTATION & RESULTS

##### A. Dataset and Tools

The MUSDB18 dataset [1] consists of 150 professionally mixed stereo tracks, each of these has an average length of between 3-5 minutes. There is an extensive variety of styles present in the dataset from multiple genres, including Pop, Rock, Jazz, Electronic, Hip-Hop. Each audio recording is sampled at 44.1 KHz, MP4 format and is converted to WAV format, allowing it to be processed by the Librosa library.

For future stages of model training and validation, additional datasets such as MedleyDB [19], Free Music Archive (FMA), and DSD100 can be considered to improve diversity and cross-genre generalization. MedleyDB [19] gives detailed instrument annotations useful for beat-specific modeling, FMA offers large collection of genre-rich open-source audio, and DSD100, used in the SiSEC evaluation campaign, contains 100 professionally mixed tracks suitable for benchmarking.

Audio preprocessing and feature extraction for EDA were done using Librosa [14], enabling computation of STFTs, chroma, MFCCs, and tempo-based features. SoundFile, PyDub, and FFmpeg were used for waveform reading, conversion, editing and manipulation. All these tools collectively provided an in-depth analysis of harmonic and percussive content within the dataset, forming a strong foundation for subsequent model development.

For exploratory analysis, fixed-length segments were processed from each recording. All subsequent explorative analysis, feature-extraction and visualization were performed using various Python libraries, including Librosa, NumPy, Matplotlib in a Google Colab environment, which provided the computational capabilities for audio processing. The experimental method was designed to create reproducible results, with consistent preprocessing steps applied across all tracks.

### B. Evaluation Metrics

The effectiveness of beat removal and music source separation models will be assessed using a combination of objective and perceptual evaluation metrics. The Signal-to-Distortion Ratio (SDR) measures the overall quality of the separated signal by comparing the reconstructed output to the original target providing an indication of how well the model retains musical content. The Signal-to-Interference Ratio (SIR) measures the model's capability to reduce unwanted components such as percussive interference with the harmonic component. Meanwhile, the Signal-to-Artifacts Ratio (SAR) verifies the level of artifacts or distortions introduced during the separation, providing insights into the naturalness and clarity of the output [15].

For baseline comparison, The range of SDR values obtained from traditional Harmonic-Percussive Source Separation (HPSS) methods, as reported in literature, is 3-5 dB for percussive suppression. In contrast, deep learning-based approaches (e.g., Demucs) produce SDR values between 6-8 dB for drum separation. This illustrates the enhanced capability of deep learning-based techniques relative to traditional signal processing techniques, when it comes to removing percussion from a mixed track.

### C. EDA and Analysis

The dataset's acoustic and spectral characteristics, whether suited for beat-removal tasks were assessed through Exploratory Data Analysis (EDA). The focus of this analysis was to evaluate the amplitude variations, frequency distributions, and correlations between the harmonic and percussive content. Fig. 3, 4, 5 & 6 show how the sample's waveform, spectrogram, correlation and fourier analysis demonstrate rhythmic intensity, frequency content and beat structure of the audio samples.

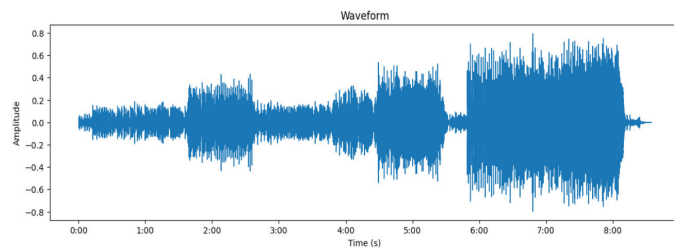


Fig. 3. Amplitude vs Time Plot

#### 1) Waveform Analysis:

The plot in Fig. 3, presents changes in audio signal amplitude over time, revealing zones of rhythmic intensity and hence, the potential for beat detection and suppression analysis. The first portion of Fig. 3 indicates moderate levels of audio signal amplitude, which represents soft or introductory musical sections. In the middle portion of the waveform, there is a noticeable rise in signal amplitudes

indicating beat-heavy sections, which continues throughout this higher amplitude area. Short silent or low-amplitude regions show pauses or transitions between musical phrases. This temporal and dynamic variations demonstrates clear structural segmentation within the audio, which confirms its suitability for beat detection, feature extraction, and training deep learning models for percussive audio removal.

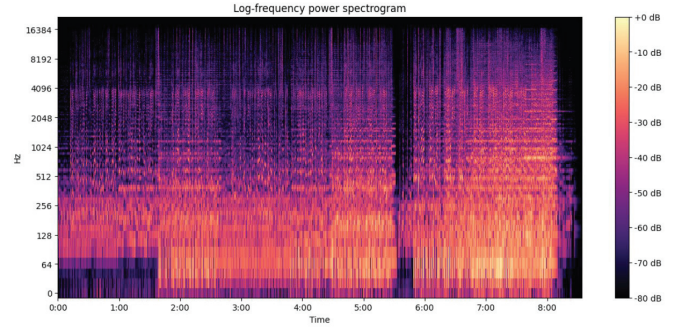


Fig. 4. Spectrogram – Frequency vs Time plot

#### 2) Spectrogram Analysis:

The log-frequency power spectrogram as shown in Fig.4 clearly indicates strong low-frequency energy concentrated below 1 kHz, corresponding to bass and percussive components, with harmonic content which extends up to 5 kHz. Repetitive vertical streaks indicate rhythmic transients, confirming a steady beat structure. Also, the distinct horizontal ridges signify the sustained melodic or vocal components. The wide dynamic range and clear separation between harmonic and percussive regions show us that the dataset effectively records both rhythmic and tonal characteristics. These characteristics confirm the suitability of the audio tracks for deep-learning based beat-removal and source-separation tasks.

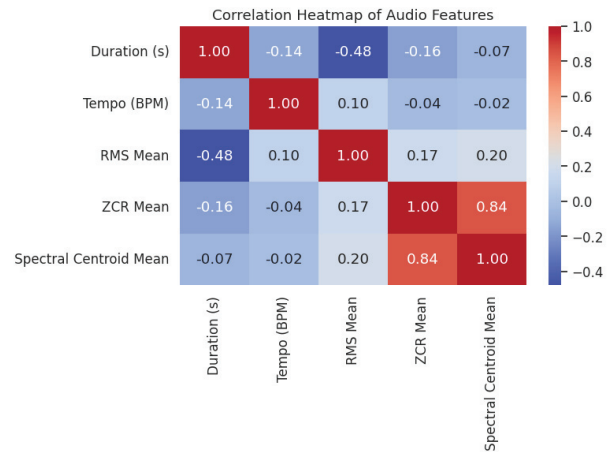


Fig. 5. Correlation Heatmap of Audio Features

#### 3) Correlation Analysis:

As shown in Fig. 5, the correlation heatmap of ZCR (zero-crossing rate) Mean and Spectral Centroid Mean displays extremely high positive correlation where ( $r = 0.84$ ) indicating the presence of higher amounts of spectral brightness in percussive sounds. RMS Mean shows moderate positive correlation with both ZCR and Spectral Centroid values, indicating that louder sound signals tend to have a higher amount of frequency content. Duration has a weak negative correlation with RMS ( $r = -0.48$ ), meaning there is a tendency for longer track durations to have lower

average levels of loudness. Tempo has little to no correlation with the other features, which shows rhythmic independence from spectral and amplitude-based attributes. Overall, these relationships indicate that energy, frequency, and percussive intensity are closely related in the dataset, which provides great info for training beat-removal models.

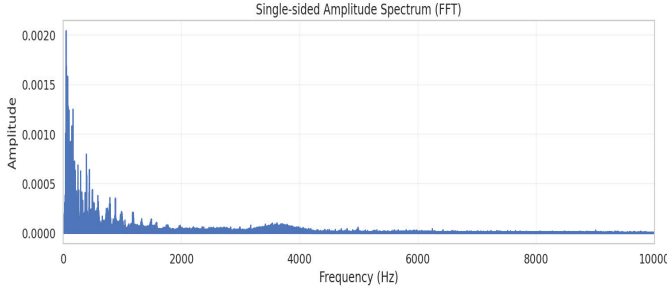


Fig. 6. Single-Sided Amplitude Spectrum (FFT)

#### 4) Fourier Transform Analysis:

The single-sided amplitude spectrum as shown in Fig. 6 indicates that there was much lower frequency content than higher frequency content within the dataset; the majority of the energy found within the dataset resides below 500 Hertz (Hz); this correlates directly to the presence of bass drums and other various low frequencies. As we move from lower frequencies to mid-frequencies (between 500 Hz and 3000 Hz), we see a gradual decrease in overall amplitude measurement, which corresponds to the increasing number of harmonic sounds being made. Also, minimal energy can be observed beyond 4000 Hz, which indicate limited high-frequency presence and a warmer tonal balance. The overall low-pass dominant spectrum shows that the audio is rhythm-heavy, with strong percussive content and moderate harmonic components. These spectral characteristics shows the suitability of the dataset for beat-removal tasks, since it clearly separates percussive elements and harmonic frequency regions, essential for effective model training.

Formulas for Feature Extraction:

Spectrogram Calculation:

$$S(t, f) = |\text{STFT}(x(t))|$$

RMS Energy:

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N x[i]^2}$$

Zero Crossing Rate:

$$\text{ZCR} = \frac{1}{N} \sum_{i=1}^{N-1} 1(x[i] \cdot x[i+1] < 0)$$

Spectral Centroid:

$$C = \frac{\sum f \cdot S(f)}{\sum S(f)}$$

#### 5) Feature Extraction Analysis:

- RMS Energy was consistently high in sections containing strong drum patterns, which indicates the presence of high percussive energy.
- Zero Crossing Rate (ZCR) increased during beat-heavy segments, showing sharp waveform transitions associated with drum hits.
- Tempo values (90–130 BPM) were detected for most tracks, confirming strong rhythmic consistency across the dataset.
- Spectral Centroid increased during drums and high-frequency percussion, reflecting the “brightness” of beat components.

#### 6) Overall Summary:

The waveform analysis indicates some clear amplitude variations, which indicates distinct musical sections and also change in dynamic intensity across the track. The spectrogram showed some strong low-frequency dominance (below 1 kHz) which are related to beats and bass, while mid and high-frequency ranges extracts the harmonic and vocal elements. The FFT amplitude spectrum shows us that most of the high energy lies in the low-frequency region (<500 Hz), which indicate the rhythmic and percussive nature of the music dataset. Statistical feature analysis has showed consistent loudness (RMS mean  $\approx 0.11$ ), moderate spectral brightness (centroid  $\approx 2.4$  kHz) and stable ZCR values, which suggests well-balanced audio quality.

The correlation heatmap showed some strong relationship between ZCR and Spectral Centroid ( $r = 0.84$ ), showing higher zero-crossing rates associated with increased spectral brightness, a typical characteristic of percussive sounds. The dataset also shows some high variations in duration and tempo, ensuring robust model generalization across different musical styles. Overall, the EDA results indicate that extracted features like (RMS, ZCR, spectral centroid, tempo) effectively differentiate the harmonic and percussive components. The analysis confirms that the dataset is acoustically diverse, rhythmically stable, and highly suitable for deep learning-based beat removal and music source separation models.

## VI. FUTURE DIRECTIONS

To overcome the gaps, several research opportunities exist for developing more specialized, efficient, and user-friendly beat removal systems:

1. Development of Annotated Beat Datasets: Creating or expanding datasets with beat time stamps, percussive event boundaries, and energy levels would facilitate supervised learning designed for beat suppression tasks.
2. Integration of Onset Detection with Deep Learning: Current onset detection algorithms, already accurate in identifying beat timings, may be used as a preprocessing step or as auxiliary training labels

during training to which enhances the percussive sensitivity in models.

3. Design of Beat-Specific Loss Functions and Metrics: New evaluation strategies which measure residual beat energy, rhythmic coherence, or beat-to-harmony ratios would be more relevant in performance feedback.
4. Cross-Genre and Cross-Tempo Model Training: Training models on various datasets that have fast, slow, polyrhythmic, and syncopated genres can be helpful in improving generalizability and robustness in the model.
5. Real-Time Beat Removal Systems: Lightweight architectures (e.g., quantized U-Net or streaming Demucs) could be optimized more for inference speed, which enables real-time beat removal for live musical performances.

## VII. CONCLUSION

The task of beat removal from music is specialized and highly relevant in karaoke generation, remixing, melody isolation and music retrieval. Traditional signal processing methods which have been efficient majorly depend on some assumptions and struggle with genre variations and variability in recording quality.

This review focuses more on deep learning as a major advancement in music source separation, with architectures such as U-Net [2], Wave-U-Net [3], and Demucs [4] showing the ability to learn hierarchical audio features which enable accurate percussive suppression while preserving melody and harmony. However, the existing research shows some key gaps and most models are optimized for general music source separation instead of beat-focused suppression. Also, there is a shortage of beat-annotated datasets and evaluation metrics that effectively measure rhythmic suppression quality.

These limitations create opportunities for developing beat-specific datasets and hybrid models which integrate beat tracking, real-time, genre-specific music systems. In summary, deep learning offers a robust and versatile framework for addressing the complex challenge of beat removal in music, which paves the way for more intelligent and context-aware audio processing systems.

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